1. Stoichiometry

1.1. Formulae

Charges of Common Ions

Name	Formula	Charges
Ammonium	NH_4^+	+1
Nitrate	NO_3^-	-1
Hydroxide	OH^-	-1
Carbonate	CO_3^{2-}	-2
Sulfate	SO_4^{2-}	-2

Molecular Formula: The number and type of different atoms in one molecule

Empirical Formula: The simplest whole number ratio of the different atoms or ions in a compound.

Molecular Formula to Empirical Formula Calculation

$$\text{Molecular Formula} = n(\text{Empirical Formula})$$

$$n = \frac{\text{Molar Mass}}{\text{Calculated Empirical Formula Mass}}$$

Element	Aluminium, Al	Oxygen, O
Mass (g)	1.08	0.96
Number of moles of atoms	$\frac{1.08}{27} = 0.04$	$\frac{0.96}{16} = 0.06$
Ratio of moles	$\frac{0.04}{0.04} = 1$	$\frac{0.06}{0.04} = 1.5$
Simplest ratio of moles	2	3

- The formula of a simple and ionic compound from the relative numbers of atoms present in a model or a diagrammatic representation

State Symbols & Chemical Equations

Balancing equations: A chemical equation is balanced when there are an equal number of atoms and charges on both sides of the equation

- **State symbols:**
  - (s) = solid
  - (l) = liquid
  - (g) = gas
  - (aq) = aqueous solution
- Names of compounds
  - A compound ending with **-ide** only contains two different elements
- A compound ending with **-ate** contains oxygen

Examples

Example 1: Simple reaction

Unbalanced equation:

$$Ca + O_2 \rightarrow CaO$$

Step 1: Count atoms on each side

Left:

- Ca = 1
- O = 2

Right:

- Ca = 1
- O = 1

Oxygen is not balanced.

Step 2: Balance oxygen

Put **2** in front of CaO:

$$Ca + O_2 \rightarrow 2CaO$$

Now count again:

Right:

- Ca = 2
- O = 2

Left:

- Ca = 1
- O = 2

Step 3: Balance calcium

Put **2** in front of Ca:

$$2Ca + O_2 \rightarrow 2CaO$$

Balanced equation:

$$2Ca + O_2 \rightarrow 2CaO$$

Example 2: Reaction with diatomic elements

Unbalanced equation:

$$H_2 + Cl_2 \rightarrow HCl$$

Step 1: Count atoms

Left:

- H = 2
- Cl = 2

Right:

- H = 1
- Cl = 1

Step 2: Balance hydrogen and chlorine

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Put **2** in front of HCl:

$H_2 + Cl_2 \rightarrow 2HCl$

Balanced equation:

$H_2 + Cl_2 \rightarrow 2HCl$

Example 3: Combustion reaction

Unbalanced equation:

$CH_4 + O_2 \rightarrow CO_2 + H_2O$

Step 1: Count atoms

Left:

- C = 1
- H = 4
- O = 2

Right:

- C = 1
- H = 2
- O = 3

Step 2: Balance hydrogen

Put **2** in front of H_2O:

$CH_4 + O_2 \rightarrow CO_2 + 2H_2O$

Step 3: Balance oxygen

Right:

Put **2** in front of O_2:

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Balanced equation:

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Example 4: Ionic compound formation

Unbalanced equation:

$Al + O_2 \rightarrow Al_2O_3$

Step 1: Count atoms

- **Left:**

  Al = 1

  O = 2
- **Right:**

  Al = 2

  O = 3

Step 2: Balance aluminium and oxygen

Al_2O_3 has **2** Al, so put **2** in front of Al

$2Al + O_2 \rightarrow Al_2O_3$

Al_2O_3 has **3 O**, O_2 has **2 O**

LCM of 2 and 3 = **6**

Put:

- **3** in front of O_2
- **2** in front of Al_2O_3

$2Al + 3O_2 \rightarrow 2Al_2O_3$

Step 3: Balance aluminium again

4 Al on the right , so put **4** in front of Al on the left

$4Al + 3O_2 \rightarrow 2Al_2O_3$

Balanced equation:

$4Al + 3O_2 \rightarrow 2Al_2O_3$

Example 5: Acid + metal reaction

Unbalanced equation:

$Zn + HCl \rightarrow ZnCl_2 + H_2$

Step 1: Count atoms

Left:

- Zn = 1
- H = 1
- Cl = 1

Right:

- Zn = 1
- H = 2
- Cl = 2

Step 2: Balance H and Cl

Put **2** in front of HCl:

$Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Balanced equation:

$Zn + 2HCl \rightarrow ZnCl_2 + H_2$

1.2. Relative Masses of Atoms and Molecules

- **Relative Atomic Mass (A_r):** the average mass of the isotopes of an element compared to 1/12th of the mass of an atom of 12C.
- **Relative Molecular Mass (M_r):** sum of relative atomic masses of all the atoms in one molecule of the compound.

1.3. The Mole and the Avogadro Constant

A **mole** of a substance is the amount that contains the same number of units as the number of carbon atoms in 12 grams of carbon-12

- A mole is the A_r or M_r expressed in grams e.g. 1 mole of Carbon-12 is equal to 12 grams.
- It is equal to 6.02×10^{23} particles; this number is called **Avogadro's constant**.
- 1 mole also occupies a volume of $24dm^3$ at room temperature and pressure

1.4. Number of Moles

$$\text{Amount of Substance} = \frac{\text{mass}}{\text{molar mass}}$$

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- Amount of Substance (mol)
- Mass (g)
- Molar Mass (Mr) in (g/mol)

1.5. Number of Moles in Aqueous Solution

$$\text{Moles} = \text{Concentration} \times \text{Volume}$$

- Moles (mol)
- Concentration (mol/dm^3)
- Volume (24 dm^3)

1.6. Moles in Gases

$$\text{Volume} = \text{No. of Moles} \times 24\text{dm}^3 \text{at r.t.p}$$

- 1 dm^3 = $1000cm^3$

1.7. Concentration

$$Concentration = \frac{\text{no. of moles}}{\text{volume}}$$

- Moles per dm^3
  - $1mol/dm^3$
- Grams per dm^3 , g/dm^3

Concentration can be measured and converted into g/dm^3 or mol/dm^3 , by multiplying the molar mass of the compound.

1.8. Moles Calculation in Acid-Base Titration

$$Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$$

20.0 cm^3 of 0.0500 mol/dm^3 HCl reacts with the 25.0 cm^3 of $Ca(OH)_2$.

Determine the concentration of $Ca(OH)_2$ in g/dm^3. Use the following steps.

- Calculate the number of moles in 20.0 cm^3 of 0.0500 mol/dm^3 HCl.

..... mol

We will use the formula:

$$\text{Number of Moles} = \frac{\text{Concentration X Volume}}{\text{1000 (change unit)}}$$

20 x 0.0500 divide 1000 = 0.001

1.9. Percentage Composition by Mass

- $\text{Percentage Composition by Mass} = \frac{\text{Mass of Element}}{\text{Molecular Mass}} \times 100$

Example Question (0620/42/F/M/23)

Calculate the percentage composition by mass of nitrogen in $(NH_4)_2SO_4$.

percentage of nitrogen = ... % [2]

Step 1: Calculate the molar mass of $(NH_4)_2SO_4$

Step 2: Find the mass of Nitrogen.

Step 3: Use the formula and multiply by 100.

Step 4: That will be the percentage of the question given.

1.10. Percentage Purity & Yield

- $\text{Percentage Purity} = \frac{\text{mass of product (pure)}}{\text{mass of compound (impure)}} \times 100$
- $\text{Percentage Yield} = \frac{\text{actual mass obtained}}{\text{calculated/theoretical mass}} \times 100$

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